

Training Real-Time Machine Learning Models to Detect and Stop Cyberattacks

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Abstract

This project focuses on how machine learning (ML) models can be trained and used to detect and stop cyberattacks in real time. Using public datasets like CICIDS2017 and UNSW NB15, several models were tested to see which ones gave the best balance between accuracy, speed, and reliability. The study used both data analysis and short interviews with cybersecurity professionals to understand what works in real settings. Four main models were tested including Decision Trees, Random Forest, Neural Networks, and Gradient Boosted Trees. The Gradient Boosted Tree model performed the best overall, reaching around 98 percent accuracy while keeping latency below 20 milliseconds. Neural Networks were slightly better at detecting new attacks but took longer to train and required more computing power. People interviewed during the project shared that explainable models and consistent retraining are important for long term success. The findings show that smaller and more efficient models can still be highly effective if they are tuned properly and checked for accuracy over time. Overall, this capstone project shows that combining adaptive and low latency ML models with intrusion detection systems can make real time cybersecurity faster and more reliable.

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Introduction

While the complexity and persistence of cyberattacks continued to grow, traditional rule-based security systems struggled to detect and respond to threats in real time. Intrusion detection systems (IDS) relied on controlled rules and were more vulnerable to sophisticated and adaptive attacks. In this setting, ML systems became a pioneering tool for cybersecurity because of their ability to identify complex patterns and anomalies across large datasets (Ali, 2025; Kantharaju, 2024). The advancements in ML algorithms, specifically deep learning, streaming analytics, and hybrid modeling, showed great promise in detecting cyberthreats with high accuracy. However, along with these benefits came difficult challenges when training models for real-time detection and prevention, such as the need for fast and continuous learning, adaptability to new attacks, and low-latency responses (Mohale, 2025).

This research aimed to investigate how new ML models were effectively trained to detect cyberattacks in real time and prevent them. It focused on the datasets, technical strategies, and frameworks that enabled real-time threat detection. Cybersecurity professionals and IT organizations continued to face pressure to stay proactive rather than reactive in their defense mechanisms. With ML's predictive capabilities, systems deployed in dynamic and high-speed networks such as IoT-based infrastructures required refined approaches to model training, data handling, and system integration.

Statement of Purpose

The purpose of this study was to explore the methods that new ML models could be trained and used to detect and stop cyberattacks in real time before any damage was done.

Research Question: How could new machine learning models be trained to detect and stop cyberattacks in real time?

This study aimed to answer this question by identifying and evaluating machine learning architectures that were suitable for real-time cybersecurity applications, including but not limited to streaming models and online learning. The second objective was to examine effective training techniques using real-time datasets such as IoT-23 and CICIDS2017. The third objective was to assess model performance in terms of speed, accuracy, false positive rate, and the ability to adapt under evolving attack scenarios. Another goal was to explore methods for continuous learning and maintaining robustness against adversarial behavior. The main goal was to understand the feasibility of these practices and determine which approach was most effective for implementing real-time, ML-based intrusion detection systems.

Literature Review

As cybersecurity threats continued to evolve and grow in complexity, many researchers began to focus on how ML could aid in the detection and prevention of these attacks in real time. Many studies went over the different ways that ML could be used in intrusion detection systems (IDS), especially in environments where privacy and efficiency were most important. This literature review went over recent approaches and explained how they worked, while also showing where there were still gaps that future research could address.

In a recent study by Ditale et al. (2025), a system called CyberMALT was introduced. The system used ML to analyze network traffic by looking at metadata instead of the actual content of the data being transmitted. This technique made the system much more efficient and helped maintain privacy when working with encrypted data. The system learned what normal traffic looked like and then flagged any data that seemed suspicious. It used anomaly detection to find these unusual patterns and classify what type of attack it was, all while

keeping the rate of false alarms low. The authors conducted a case study using a real-time network simulation to test CyberMALT's accuracy and privacy measures. CyberMALT was an important example to this research because it showed that ML could be applied in real time while maintaining privacy.

Federated Learning Models

Another area that had gained attention was federated learning (FL). FL was a method that allowed ML models to be trained on devices locally instead of sending all the data to a central server. Ghimire and Rawat (2022) conducted a simulation-based study using a virtual IoT environment to evaluate how FL helped secure Internet of Things (IoT) devices by detecting attacks such as denial of service (DoS) and intrusions. Their research showed that FL greatly protected privacy and limited data sharing, which was extremely important for systems with sensitive information. However, the challenges they pointed out were not simple, as FL still faced cyber threats such as data poisoning. As a countermeasure, the authors suggested that encryption and blockchain could help secure federated learning.

Yang et al. (2019) provided background on how federated learning worked. Their paper went over different types of FL, such as cross-device and cross-organization learning. The authors discussed how FL helped protect user privacy while addressing the problem of "data islands," where important data remained isolated in one location. They also discussed how secure computation and encryption improved the safety of FL systems. Although their work focused more on the concept than real-time use, it helped provide understanding of why federated learning was useful in cybersecurity. Their paper served as a conceptual analysis rather than an experimental or implementation-based study.

Threat Detection ML Models

Some researchers focused on which ML models and deep learning methods worked best for threat detection. Xin et al. (2018) reviewed common techniques used in IDS, including signature-based, anomaly-based, and hybrid approaches. The authors conducted a comparative literature review and benchmarked performance statistics from various studies, showing that deep learning models such as neural networks were strong at solving complex attack patterns. However, these models required significant computing power and large datasets. This study was helpful because it compared different methods and showed the trade-offs between speed and accuracy.

Ahsan et al. (2022) focused on how ML was used to prevent common cyber threats such as DoS attacks, phishing, and malware. The authors conducted a systematic review of 100 peer-reviewed studies, analyzing the effectiveness of various ML models in detecting and mitigating different types of cyber threats in multiple environments. This review showed that ML was effective at detecting unusual behaviors, such as someone logging in from an unfamiliar location at an unusual time. They also noted that as attackers' tactics changed, ML models needed to be updated regularly to keep up. This article supported the idea that ML models must be able to adapt quickly to new attacks in real time.

This study focused on how new ML models could be trained to find cyberattacks and stop them before they happened in real time. It explored tools that could help such as federated learning, real-time anomaly detection, and secure data analysis. The research question for this study was: **“How can new machine learning models be trained to detect and stop cyberattacks in real time?”**.

Methodology

Study Design

This study implemented a mixed methods approach that combined quantitative data analysis with qualitative information to better understand how new machine learning (ML) models could be trained to detect and prevent cyberattacks in real time. The quantitative component followed a secondary data analysis design using existing cybersecurity datasets, while the qualitative component included interviews with cybersecurity professionals and IT staff who provided insight from their experience in the field. The overall design of the research was quasi-experimental because it compared the performance of several ML models without directly manipulating any variables.

Procedures

The primary goal of this study was to examine and compare the performance of different ML models in detecting and mitigating cyber threats in real time. The key metrics used for evaluating model performance included accuracy, precision, recall, and F1 score (a metric that combines both precision and recall). These measures helped determine which models such as Decision Trees, Neural Networks, and Random Forests were the most effective in dynamic and rapidly changing threat environments. The models were trained and tested under similar conditions to ensure fair comparison across different algorithm types.

Sampling/Data Sources

This research used two publicly available, labeled datasets that have been widely adopted in cybersecurity research:

UNSW-NB15 – A modern dataset that contained diverse attack scenarios including backdoors, worms, and denial of service (DoS) attacks.

CICIDS-2017 – A comprehensive dataset that included both real and simulated malicious traffic collected from a controlled network environment.

Both datasets were anonymized and ethically sourced. Data cleaning and feature selection were conducted to optimize model training and improve efficiency. Examples of the extracted features included packet flow duration, protocol type, source and destination IP patterns, and port behaviors. Stratified sampling was used to maintain balanced representation of different attack types. Standard data preprocessing techniques such as normalization and encoding were applied using Python libraries including Scikit-learn and Pandas.

Qualitative Aspect

In addition to data analysis, the study included interviews with cybersecurity professionals and IT staff to collect additional qualitative insights. A purposive sampling approach was used to identify participants based on their experience with machine learning or cybersecurity and their willingness to participate. A total of five professionals were invited through professional networking platforms and email. Each participant provided informed consent after reviewing the study's purpose and how their responses would be used. Interview questions focused on participants' experience using ML in real-time threat detection and prevention, challenges they had encountered, and their opinions about model effectiveness and reliability.

Data Collection/Storage

For the quantitative portion, the datasets were downloaded directly from their official sources and stored locally in CSV format. All processing and analysis were performed using Python in a secure computing environment.

For the qualitative portion, interview data were collected through online video calls or live chat messages. Audio and text transcripts were recorded and stored locally in password-protected folders. Only the researcher had access to these files, and no identifying information was included in the analysis.

Analysis Methods

For the quantitative analysis, the ML models were trained and tested using a k-fold cross-validation method to ensure consistent evaluation across multiple data partitions. Performance metrics such as accuracy, precision, recall, and F1 score were used to measure each model's effectiveness. Libraries including Scikit-learn, TensorFlow, and Pandas supported the model training and evaluation process.

For the qualitative analysis, interview transcripts were reviewed and coded manually to identify recurring ideas and opinions. The data were organized into general themes such as model flexibility, implementation difficulty, and overall trust in AI systems. Subtopics were created when needed to capture more specific insights. These themes helped explain real-world challenges faced when implementing ML systems for cybersecurity purposes and provided context to the quantitative findings.

Note - See Appendix #1 for Interview Questions

Results

Introduction

This section presents the results of the study, which analyzed both quantitative and qualitative data collected to explore how machine learning models can detect and stop cyberattacks in real time. The quantitative results came from model testing and evaluation using the CICIDS2017 and UNSW-NB15 datasets, while the qualitative results came from

interviews with cybersecurity professionals who shared their perspectives on the use of ML in real-time threat detection. Data were analyzed using accuracy, precision, recall, and F1-score metrics for the quantitative portion, and recurring themes were identified for the qualitative portion.

Quantitative Findings

The primary goal of the quantitative analysis was to compare the performance of four machine learning models that are Decision Tree, Random Forest, Neural Network, and Gradient Boosted Trees when trained and tested on two public datasets representing real-world cyberattacks. This comparison was designed to find out not only which model achieved the highest accuracy, but also how each model behaved across different types of attacks, levels of traffic imbalance, and real-time demands. The analysis focused on evaluating precision, recall, F1 score, and false positive rates to better understand how consistently each model identified both common and unique intrusions. In addition, the comparison helped reveal how differences in the structure of the datasets, such as the presence of more modern attacks in the UNSW-NB15 and traffic pattern separations in CICIDS2017, changed the overall model performances. These metrics made it possible to gauge the strengths and limitations of each model in realistic cybersecurity settings/conditions rather than relying on accuracy alone.

Model Performance Overview

Table 1 summarizes the accuracy, precision, recall, F1 scores, and false positive rates for all four models across both datasets. The table shows that Gradient Boosted Trees achieved the highest overall accuracy on the CICIDS2017 dataset at 98 percent, with Random Forest and the Neural Network following closely behind. Decision Trees scored lower in all metrics, especially in false positive rate. For the UNSW NB15 dataset, a similar pattern was

prevalent, with Gradient Boosted Trees again having the best overall balance of metrics. The table also demonstrates that Neural Networks had slightly higher recall across both datasets, which suggests that they were more sensitive to detecting unique attacks. These results point to Gradient Boosted Trees being consistently strong across all categories, while Neural Networks performed well in recall but struggled with efficiency and stability.

Table 1 - Performance Metrics Across Models and Datasets

Model	Dataset	Accuracy	Precision	Recall	F1-Score	False Positive Rate
Decision Tree	CICIDS2017	0.94	0.93	0.95	0.94	0.015
Random Forest	CICIDS2017	0.96	0.97	0.96	0.96	0.010
Gradient Boosted Trees	CICIDS2017	0.98	0.99	0.98	0.98	0.006
Neural Network	CICIDS2017	0.97	0.98	0.99	0.98	0.008
Decision Tree	UNSW-NB15	0.92	0.93	0.91	0.92	0.020
Random Forest	UNSW-NB15	0.95	0.96	0.94	0.95	0.013
Gradient Boosted Trees	UNSW-NB15	0.97	0.98	0.96	0.97	0.009

Neural Network	UNSW- NB15	0.96	0.97	0.98	0.97	0.010
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Latency and Efficiency

Model efficiency was measured in milliseconds per classification. The Gradient Boosted Tree and Random Forest models processed traffic significantly faster than the Neural Network model. The Decision Tree model had the lowest latency overall but also slightly lower accuracy. These latency patterns help explain the practical differences in how each model would perform within a real environment. The low latency of Decision Trees would allow extremely fast responses, but the tradeoff in precision and the higher rate of false alerts limits its usefulness. In contrast, the moderate latency of models such as Random Forest and Gradient Boosted Trees shows that they can process traffic quickly enough to meet real time demands while still providing stronger accuracy and stability. The results in Table 2 further show that the ninety-fifth percentile latency for Gradient Boosted Trees remained within a predictable range, this indicates consistent performance even under heavier network loads. Neural Networks, while powerful in detecting unique attacks, demonstrated too much delay in their model to be relied on as the primary real time model. Overall, the latency results support the conclusion that ensemble models (which combine multiple individual models to improve overall predictive performance) deliver the most efficient balance between detection strength and system responsiveness.

Table 2 - Average Latency Per Model

Model	Average Latency (ms)	95th Percentile Latency (ms)
Decision Tree	8.5	13.2
Random Forest	14.6	23.5

Gradient Boosted Trees	18.9	28.1
Neural Network	30.4	46.2

Model Comparison Summary

Overall, the Gradient Boosted Trees model achieved the highest balance between accuracy and real-time efficiency. The Neural Network model provided strong recall and slightly higher sensitivity to new attack types but was more resource intensive. The Decision Tree model, while faster, produced more false positives compared to ensemble-based approaches.

Qualitative Findings

The qualitative portion of this study included interviews with five cybersecurity professionals who discussed their experience using machine learning in real-time intrusion detection. Analysis of the interview transcripts revealed three main themes: model interpretability, operational trust, and ongoing maintenance requirements.

Theme 1: Model Interpretability

Model interpretability was one of the strongest and most consistent priorities mentioned by the interview participants and each interviewee explained that a model is only useful if analysts can understand why it decides, especially in environments with high pressure and where alerts must be verified quickly. One participant stated, “If I cannot explain why an alert happened, then I cannot trust it enough to act on it,” which highlighted how interpretability affects the decisions analysts make in real time. A participant described a situation where a model produced an alert with no explanation and caused delays because analysts had to manually trace the traffic to determine whether the alert was valid.

This concern aligned closely with the quantitative findings where Gradient Boosted Trees and Decision Trees both provide clear decision pathways that can be reviewed by analysts and match participants' preferences. This shows that interpretability is not simply a feature preference but an actually critical requirement for real time intrusion detection systems.

Theme 2: Operational Trust and Alert Management

Some participants discussed alert fatigue and the importance of minimizing false positives. They preferred models that maintained a low false-positive rate even if it meant slightly reducing sensitivity. Another participant said, "I'd rather miss one or two minor alerts than chase a hundred false ones a day."

Gradient Boosted Trees, which produced the lowest false positive rates in both datasets, best matched participants' expectations for a trustworthy model. Neural Networks performed well in detecting more unique attacks but sometimes generated more unnecessary alerts, which participants would consider unacceptable in a real time setting. Decision Trees also had a higher rate of false positives which made them impractical even though they were the fastest model. Random Forest models were viewed more positively because they offered a stable and consistent pattern of predictions.

Theme 3: Continuous Model Updating

The final major theme was the need for continuous updating and retraining of machine learning models where all the participants agreed that cyberattacks evolve rapidly and that any machine learning model left unchanged for long periods will lose effectiveness. One participant said, "If the model is not updated often, it becomes useless in a month," this demonstrates how quickly new attack methods can cause models to fall behind. Participants described the challenge of integrating new patterns of malicious behavior, explaining that

threat actors often modify their techniques many different times to bypass existing detection rules.

This theme directly supported the quantitative findings regarding model adaptability and performance over time and models like Gradient Boosted Trees and Random Forests showed a stronger ability to adapt to new attack types, but even these models require periodic retraining to stay accurate and not fall behind. Neural Networks, while they are strong in pattern recognition, depend heavily on current and representative data which can make them a less useful choice. This theme reinforces the idea that a successful real time system requires both strong training initially and ongoing maintenance to stay effective against evolving threats.

Summary of Findings

The results showed that the Gradient Boosted Trees model achieved the best overall performance across both datasets, maintaining high accuracy and reasonable latency. The Neural Network model produced the highest recall but required more computing resources. Qualitative findings supported these quantitative results by highlighting the importance of model transparency, reliability, and adaptability in real-time cybersecurity environments. These results provide a factual basis for the interpretation and analysis that will be discussed in the following section.

Discussion

Results Summary and Research Question

This study focused on finding how machine learning (ML) models could be trained to detect and stop cyberattacks in real time. The main research question asked how new ML models can be trained to detect and stop cyberattacks in real time. The results showed that the Gradient Boosted Trees (GBT) model performed the best overall by keeping high accuracy

and low latency. The GBT model reached about 98 percent accuracy on the CICIDS2017 dataset and 97 percent on the UNSW-NB15 dataset while maintaining quick response times under 20 milliseconds. Neural Networks had strong recall but took longer to train and needed more computing power. Decision Tree and Random Forest models still performed well but were not as effective in fast, changing network conditions.

The interviews supported these results and helped explain what mattered most to people working in cybersecurity. Three main priorities stood out: model interpretability, operational trust, and continuous model updates. The participants preferred models that were easier to understand and explain, especially when they needed to react quickly to alerts. Overall, the study supported the research question by showing that ML models can detect and prevent attacks in real time when they are trained efficiently and regularly updated.

Interpretation of Results and Implications

The results showed that machine learning can make a major difference in modern cybersecurity if the models are built and trained in the right way. The Gradient Boosted Trees model proved that high accuracy does not always require a complicated design. It showed that simpler and lighter models can still perform at the same level as deep learning networks while running faster and using fewer resources.

The main takeaway was that the balance between accuracy and speed is what makes a real-time model work. A model that detects an attack but takes too long to react is not useful in live environments. The Neural Network model was an example of this trade-off as it performed well but slowed down response times. The interviews confirmed that speed and trust were more important than pure accuracy. People wanted models that gave clear reasons for their alerts instead of “black box” systems that were hard to explain.

This study's results also pointed to an important practical lesson: ML systems in cybersecurity cannot stay static. The data used to train them eventually becomes outdated as attacks change over time. To stay effective, models need scheduled retraining and consistent monitoring. In practice, this means that security teams could retrain models every few weeks or monthly to keep accuracy levels high and false positives low.

Comparison with Existing Literature

The results of this study were similar to what other researchers have found. For example, Ditale et al. (2025) introduced the CyberMALT system, which used ML to detect abnormal network traffic in real time while protecting data privacy. Like CyberMALT, this project showed that it is possible to use machine learning for fast and private detection without sacrificing accuracy. Xin et al. (2018) and Ahsan et al. (2022) also found that ML-based intrusion detection systems performed better than older rule-based ones, especially when dealing with large datasets and complex attacks.

However, this project found that deep learning models are not always the best choice for real-time defense. While earlier research by Nguyen et al. (2022) and Alshamrani et al. (2020) showed deep learning models reaching very high accuracy, this study revealed that they are not as efficient when time and resources are limited. This supports what Shen et al. (2023) argued about focusing more on explainability and efficiency instead of chasing perfect accuracy.

This project also added something new to the conversation by including professional insights. While most papers focused only on model design and numbers, the interviews here showed what actually matters in practice are models that are easy to explain, fast to run, and simple to retrain. That connection between technical results and real-world usability helps fill

a gap in the existing research and shows how to make ML security tools more realistic for everyday use.

Limitations

Even though the results were strong, this study had a few limitations. The datasets that were used, CICIDS2017 and UNSW-NB15, were public and simulated. This means they did not completely represent the complex and encrypted traffic that happens on real networks. Because of this, model performance in a live setting might differ slightly. The testing environment was also controlled, so network lag and hardware limits were not as unpredictable as they would be in an actual organization.

Another limitation was the small number of interview participants. Only five cybersecurity professionals were interviewed, which gave helpful insights but not a wide range of experiences. Their answers might not represent the full variety of opinions across different industries. Finally, this study focused on supervised learning models and did not test other growing approaches like federated learning or reinforcement learning, which could also be useful for real-time defense.

Conclusion and Recommendations

In conclusion, this study found that new ML models can detect and stop cyberattacks in real time when they are designed to be both fast and reliable. The Gradient Boosted Trees model offered the best overall balance between accuracy, speed, and ease of use. The study also showed that cybersecurity tools should not only be accurate but understandable to the people using them. Analysts need to trust what the model is doing to respond confidently to threats.

Based on these findings, several recommendations can be made:

- Use lightweight models first and start with simpler and faster models like Gradient Boosted Trees before moving to complex deep learning systems.
- Keeping retraining models as regular updates is key to maintaining accuracy as new attacks appear.
- Improve explainability because future ML security tools should show why an alert is triggered so analysts can verify it quickly.
- Explore adaptive learning so that future research can focus on models that automatically update themselves based on new patterns without full retraining.

This study investigated how machine learning models can be trained and used to detect and stop cyberattacks in real time and the project combined analysis of two large public datasets with interviews from cybersecurity professionals to evaluate model accuracy, reliability, latency, and operational usability. The results showed that Gradient Boosted Trees offer the best balance between their speed, interpretability, adaptability, and accuracy. Neural Networks have advantages in detecting unique or emerging threats, but they are slower and harder to interpret in high pressure environments. Decision Trees and Random Forest models each provide benefits but do not match up to the overall performance of Gradient Boosted Trees. These findings demonstrate that machine learning can significantly improve the speed and quality of intrusion detection when models are well-tuned, retrained regularly, and deployed with consideration for their strengths and limitations. This study contributes to the growing field of intelligent security systems and highlights the practical guidelines for implementing a machine learning based defense in real world networks.

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Appendix

#1 - Interview Questions

1. Can you describe your experience using machine learning in real-time cybersecurity threat detection?
2. What challenges have you faced when implementing ML-based intrusion detection systems in the live environment?
3. How do you assess the effectiveness of an ML model in preventing or mitigating cyberattacks?
4. In your opinion, how frequently should real-time ML models be retrained or updated to stay effective?
5. Do you think federated learning is practical for cybersecurity applications in your organization? Why or why not?
6. How important is model transparency and explainability in your decision to use ML for security?
7. Do you believe AI-powered models are trustworthy enough to operate independently in detecting and stopping threats? Why or why not?